

William Ma

(240) 283-7288 · williamma222@gmail.com · Potomac, MD

EDUCATION

- **Duke University · Pratt School of Engineering** 2026 - Present
 - B.S.E. Mechanical Engineering: emphasis on robotics, autonomous systems, and systems integration
- **Winston Churchill High School**
 - GPA: 3.98/4.00 · WGPA: 4.95/5.00 · National Merit Finalist 2023 - 2026

RESEARCH & PROFESSIONAL EXPERIENCE

- **CAD Mentor – FTC Team #30387 (PAID)** 2026 - Present
 - Mentor competition robotics students in CAD, mechanical design, prototyping, and iteration.
- **Binghamton University — Robotics Research** 2025
 - Investigated large-language-model approaches to multi-robot route control and coordination using LiDAR-derived environmental data
- **Junior Swim Coach — Potomac Glen Summer Swim Team (PAID)** 2024-2025
 - Paid coach supporting instruction and practices for swimmers across age groups on a 100+ member team

ENGINEERING & TECHNOLOGY PROJECTS

- **3-Wheel Modular Swerve Robot · *Independent*** 2026 - Present
 - Designing and fabricating a 3-pod equilateral swerve chassis for close-to-full holonomic freedom. Co-axial design with an emphasis on cost-effective engineering and full modularity with an attachable CAN Bus.
 - Uses a Raspberry Pi 4B for a high-level computer with a slaved Raspberry Pi 2040 microcontroller for real time PID control. Webcam and 6 IR distance sensors allow for AI pathing and reactive motor control.
- **First Tech Challenge (FTC) Team #24028 · *Founder & Captain · Design Award · Innovate Award*** 2023-2026
 - Solo design and fabrication of a 33-lb competition robot with 150+ custom components; integrated CAD, 3D printing, drivetrain, turret, flywheel, sorting, odometry, vision systems, and a fully custom compact linear actuator. Led the team in previous years through assembly and design.
 - Developed Java autonomous software using PID control, wheel odometry/localization, and vision-assisted targeting; iterated mechanical and software systems through competition testing.
- **UMD Jumpstart – JARVIS Sort · *Hardware Lead*** 2025
 - Developed the hardware system for an AI-powered sorting robot using a Core-XY gantry, Raspberry Pi, stepper motors, motor drivers, and a 5-DOF robotic claw.
 - Integrated mechanical and electrical subsystems for automated object handling, gaining hands-on experience across robotics hardware and system integration.
- **Science Olympiad – Electric Vehicle · *Independent · #3 In State*** 2025-2026
 - Built an Arduino-controlled competition EV with a 4-bar linkage system, DC motors, and a sprung odometry pod designed to maintain wheel contact for more consistent position tracking
 - Iterated drivetrain geometry, packaging, and control under strict power, size, and weight constraints; earned 1st at regional and 3rd at state competition.
- **Science Olympiad – Robot Tour · *Independent · #1 In Region · #2 In State*** 2025-2026
 - Developed an open-loop differential-drive robot using a belt-driven stepper system and custom software synchronization to coordinate motors without closed-loop feedback.
 - Designed the transmission and control architecture around competition constraints, emphasizing repeatable motion and low-cost fabrication.
- **Science Olympiad – Tower & Bridge · *Independent · #1 In Region · #6 In State*** 2023-2025
 - Designed and fabricated optimized balsa structures; achieved 2,000+ Tower efficiency and 1,000+ Bridge efficiency through load-path, truss, joint, and material-placement optimization

TECHNICAL SKILLS

- **Robotics-Related Skills**
 - OnShape, KiCAD, 3D Printing, ROS2, Gazebo, PID, Odometry, Sensor Fusion, Mechanical Assembly, PCB Design, Java, Python, C/C++, JavaScript, Arduino, Raspberry Pi, I²C, SPI, UART
- **Misc Skills**
 - Visual Arts · 4x Scholastic Gold Key, 2x Silver Key, 3x HM; Tennis · Churchill Boy's Varsity Captain & 2x County Champion